LangChain, LangGraph & LangSmith - Complete Guide

Master Index & Learning Path

Total Concepts Covered: 270+
Learning Time: 30-50 hours
Implementation: ai-langx/ expense tracker

📚 Documentation Structure

Part 1: Foundations

File: `LANGCHAIN_LANGGRAPH_LANGSMITH_COMPLETE_GUIDE.md`
Concepts: 40+
Reading Time: 3-4 hours

Chapters:

1. Prerequisites & Environment Setup
2. JavaScript/Node.js Essentials
3. LLM Fundamentals
4. Understanding AI Orchestration
5. Models - Talking to LLMs
6. Prompts - Instructing LLMs

What You'll Learn:

- Development environment setup
- JavaScript async/await, Promises, ES6 modules
- LLM basics: tokens, temperature, context windows
- How LangChain/LangGraph/LangSmith work together
- ChatOpenAI usage and configuration
- Prompt templates and best practices

Start Here: ☑ Read this first if you're new to AI/LangChain

Part 2: LangChain Core

File: `PART_2_LANGCHAIN_CORE.md`
Concepts: 60+
Reading Time: 5-6 hours

Chapters: 7. Tools - Extending LLM Capabilities 8. Agents - Autonomous Decision Making 9. Chains - Sequential Operations 10. Memory - Conversation Context

What You'll Learn:

- StructuredTool creation and Zod validation
- AgentExecutor and tool-calling loops
- ReAct, OpenAI Functions, and Conversational agents
- LLMChain, SequentialChain, RetrievalQAChain
- ConversationBufferMemory and variants
- Real examples from ai-langx/ implementation

Read After: Part 1

Part 3: LangChain RAG Pipeline

File: `PART_3_LANGCHAIN_RAG.md`
Concepts: 50+
Reading Time: 4-5 hours

Chapters: 11. Document Loaders - Ingesting Data 12. Text Splitters - Chunking Documents 13. Embeddings - Vector Representations 14. Vector Stores - Storing Embeddings 15. Retrievers - Finding Relevant Info 16. RAG Chains - Question Answering

What You'll Learn:

- PDFLoader, CSVLoader, TextLoader
- RecursiveCharacterTextSplitter strategies
- OpenAIEmbeddings and similarity calculation
- MemoryVectorStore and persistence
- Retriever types and filtering
- Complete RAG implementation from ai-langx/

Read After: Part 2

Part 4: LangChain Advanced

File: `PART_4_LANGCHAIN_ADVANCED.md`
Concepts: 50+
Reading Time: 4-5 hours

Chapters: 17. LCEL - Expression Language 18. Runnables - Building Blocks 19. Output Parsers - Structured Responses 20. Callbacks - Event Handling 21. Caching - Performance Optimization 22. Advanced Retrievers 23. Advanced Chains

What You'll Learn:

- RunnablePassthrough, RunnableLambda, pipe()
- StructuredOutputParser with Zod
- Custom callbacks for logging/metrics
- Redis/InMemory caching
- Multi-Query, Ensemble, Parent Document retrievers
- Summarization, SQL, API, Router chains
- How to leverage these in ai-langx/

Read After: Part 3

Part 5: LangGraph Workflows

File: `PART_5_LANGGRAPH.md`
Concepts: 40+
Reading Time: 4-5 hours

Chapters: 24. StateGraph Fundamentals 25. Nodes - State Transformers 26. Edges - Connecting Flow 27. State Management 28. Conditional Routing 29. Parallel Execution 30. Persistence & Checkpoints 31. Streaming & Real-time Updates 32. Advanced LangGraph Patterns

What You'll Learn:

- StateGraph creation and compilation
- Node functions and state updates
- Simple, conditional, and parallel edges
- Zod state schemas
- START, END, and routing logic
- SQLite/Postgres checkpointers
- Time travel and branching
- MessageGraph and prebuilt agents
- Real workflows from ai-langx/ (IntentRouter, Reconciliation)

Read After: Part 2 or 3

Part 6: LangSmith Observability

File: `PART_6_LANGSMITH.md`
Concepts: 40+
Reading Time: 3-4 hours

Chapters: 33. Observability Basics 34. Tracing & Debugging 35. Tags & Metadata 36. Analytics & Dashboards 37. Datasets & Testing 38. Experiments & A/B Testing 39. Evaluators & Quality Assurance 40. Production Monitoring

What You'll Learn:

- Automatic tracing setup
- Trace explorer and filtering
- Performance metrics (latency, cost, tokens)
- Creating test datasets
- Running experiments
- LLM-as-judge evaluators
- Alerting and error tracking
- Real dashboard usage for ai-langx/

Read After: Part 2 (can read anytime)

Part 7: Integration & Production

File: `PART_7_INTEGRATION.md`
Concepts: 30+
Reading Time: 3-4 hours

Chapters: 41. Integrating All Three Frameworks 42. AI-LangX Implementation Walkthrough 43. Production Best Practices 44. Troubleshooting & Debugging

What You'll Learn:

- How LangChain + LangGraph + LangSmith work together
- Complete code walkthrough of ai-langx/
- Request flow from frontend to backend
- Error handling patterns
- Performance optimization
- Security considerations
- Deployment strategies
- Common issues and solutions

Read After: All previous parts

🎯 Learning Paths

Path 1: Complete Beginner (Recommended)

Read in order: Part 1 → 2 → 3 → 5 → 6 → 7 → 4 (advanced last)

Total Time: 30-35 hours

Milestone Checkpoints:

- ☑ After Part 1: Understand LLM basics and setup
- ☑ After Part 2: Build simple agent with tools
- ☑ After Part 3: Implement basic RAG system
- ☑ After Part 5: Create workflow with StateGraph
- ☑ After Part 6: Set up LangSmith tracing
- ☑ After Part 7: Understand complete ai-langx/ codebase
- ☑ After Part 4: Master advanced patterns

Path 2: Experienced Developer (Fast Track)

Read: Part 1 (skim) → Part 2 (detailed) → Part 5 → Part 3 → Part 6 → Part 7 → Part 4

Total Time: 20-25 hours

Focus Areas:

- Tools and Agents (Part 2)
- LangGraph workflows (Part 5)
- RAG implementation (Part 3)

- Production practices (Part 7)

Path 3: Focus on RAG

Read: Part 1 → Part 3 (detailed) → Part 4 (retrievers section) → Part 2 (chains) → Part 6

Total Time: 15-20 hours

For: Document Q&A systems, knowledge bases, search

Path 4: Focus on Agents

Read: Part 1 → Part 2 (detailed) → Part 5 → Part 4 (LCEL section) → Part 6

Total Time: 15-20 hours

For: Autonomous systems, tool-calling applications

Path 5: Focus on Workflows

Read: Part 1 → Part 5 (detailed) → Part 2 (agents) → Part 6 → Part 7

Total Time: 15-18 hours

For: Multi-step processes, complex orchestration

📖 Quick Reference by Concept

Models & Prompts

- ChatOpenAI: Part 1, Chapter 5
- PromptTemplate: Part 1, Chapter 6
- Temperature & tokens: Part 1, Chapter 3
- Streaming: Part 1, Chapter 5

Tools & Agents

- StructuredTool: Part 2, Chapter 7
- AgentExecutor: Part 2, Chapter 8
- Tool-calling loop: Part 2, Chapter 8
- Zod validation: Part 2, Chapter 7

RAG Pipeline

- Document loaders: Part 3, Chapter 11
- Text splitters: Part 3, Chapter 12
- Embeddings: Part 3, Chapter 13
- Vector stores: Part 3, Chapter 14
- Retrievers: Part 3, Chapter 15
- RetrievalQAChain: Part 3, Chapter 16
- Advanced retrievers: Part 4, Chapter 22

Chains

- LLMChain: Part 2, Chapter 9
- SequentialChain: Part 2, Chapter 9
- RetrievalQAChain: Part 3, Chapter 16
- Summarization chains: Part 4, Chapter 23
- SQL/API chains: Part 4, Chapter 23

Memory

- ConversationBufferMemory: Part 2, Chapter 10
- WindowMemory: Part 2, Chapter 10
- SummaryMemory: Part 2, Chapter 10

LCEL & Runnables

- Pipe operator: Part 4, Chapter 17
- RunnablePassthrough: Part 4, Chapter 18
- RunnableLambda: Part 4, Chapter 18
- RunnableBranch: Part 4, Chapter 18

LangGraph

- StateGraph: Part 5, Chapter 24
- Nodes & edges: Part 5, Chapters 25-26
- State management: Part 5, Chapter 27
- Conditional routing: Part 5, Chapter 28
- Checkpoints: Part 5, Chapter 30
- Streaming: Part 5, Chapter 31

LangSmith

- Tracing setup: Part 6, Chapter 33
- Tags & metadata: Part 6, Chapter 35
- Dashboards: Part 6, Chapter 36
- Datasets: Part 6, Chapter 37
- Experiments: Part 6, Chapter 38
- Evaluators: Part 6, Chapter 39

AI-LangX Implementation

- Intent routing: Part 7, Chapter 42
- Expense agent: Part 7, Chapter 42
- RAG Q&A: Part 7, Chapter 42
- Reconciliation: Part 7, Chapter 42

🔧 Hands-On Exercises

Each part includes exercises:

Part 1: Set up environment, run first LLM call **Part 2**: Build expense tool, create simple agent **Part 3**: Implement PDF Q&A system **Part 4**: Add caching, create custom retriever **Part 5**: Build intent router workflow **Part 6**: Set up LangSmith, analyze traces **Part 7**: Deploy ai-langx/, run production tests

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📊 Concept Coverage

Framework	Concepts	Parts Covered
LangChain Core	110	Parts 1, 2, 4
LangChain RAG	50	Part 3, 4
LangGraph	40	Part 5
LangSmith	40	Part 6
Integration	30	Part 7
Total	**270+**	**All Parts**

🎓 Completion Certificate

After completing all parts and exercises:

1. ☑ Build a complete AI application using all 3 frameworks
2. ☑ Implement RAG pipeline with custom retrievers
3. ☑ Create multi-step workflows with LangGraph
4. ☑ Set up production monitoring with LangSmith
5. ☑ Debug and optimize AI applications
6. ☑ Understand ai-langx/ codebase completely

Recommended Project: Build your own version of ai-langx/ with custom features

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📝 Additional Resources

Official Documentation

- LangChain JS: https://js.langchain.com/docs
- LangGraph: https://langchain-ai.github.io/langgraphjs/
- LangSmith: https://docs.smith.langchain.com

AI-LangX Repository

- Code: `ai-langx/` folder
- Architecture: `notes/AI_LANGX_ARCHITECTURE_PLAN.md`
- Deep Dive: `notes/AI_LANGX_ARCHITECTURE_DEEP_DIVE.md`
- Mapping: `notes/AI_FRAMEWORK_MAPPING.md`

Community

- Discord: LangChain community
- GitHub: langchain-ai/langchainjs
- Twitter: @langchainai

🚀 Getting Started

1. **Set up environment** (Part 1, Chapter 1)
2. **Choose your learning path** (see above)
3. **Read parts in order**
4. **Complete exercises** at the end of each part
5. **Build projects** to practice concepts
6. **Refer back** as needed (use index above)

💡 Tips for Success

Do:

- ☑ Code along with examples
- ☑ Experiment with different parameters
- ☑ Use LangSmith to observe execution
- ☑ Read ai-langx/ code while learning
- ☑ Take breaks between parts

Don't:

- ❌ Skip fundamentals (Part 1 essential)
- ❌ Rush through without coding
- ❌ Ignore errors (debug and understand)
- ❌ Skip exercises (practice is key)

Ready to start? → Open `LANGCHAIN_LANGGRAPH_LANGSMITH_COMPLETE_GUIDE.md` (Part 1)

Questions? → Check Part 7, Chapter 44 (Troubleshooting)

Stuck? → Review relevant concept in Quick Reference above

Document Version: 1.0
Last Updated: February 9, 2026
Total Pages: ~400 (when converted to PDF across all parts)